

TECHNICAL APPROACHES

Architectural Engineering, AI Verification, Algorithm Rules & Evaluator Defense Guide

Project: DRISHTI (Disaster Response)
Document Type: Technical Report & Viva Defense
Architecture: Offline-First PWA • Edge AI

Executive Summary: This technical report details the engineering design, artificial intelligence mechanisms, rule-based decision thresholds, and API integrations powering the DRISHTI Disaster Management Platform. It is specifically prepared to answer evaluators' core architectural questions: *What technology is used, how do algorithms calculate outputs, which APIs are integrated, and why these models were selected over conventional alternatives.*

1.0 CITIZEN REPORTING & MULTI-MODAL AI VERIFICATION

MODULE 1 • REPORT SECTION

Q1 In the Report section, which AI is used, how does the algorithm verify fake vs genuine reports, which API is used, and why this model instead of heavy LLMs?

MODEL / AI USED

Rule-Based NLP Agent + In-Browser Computer Vision (Byte Entropy)

APIS & STORAGE

HTML5 Geolocation API, Canvas 2D API, IndexedDB v2

⚙️ How the Verification Algorithm Works (Step-by-Step Rules):

- **1. Instant Fake Filter:** Description me agar prank words ('test', 'prank', 'joke', 'alien', 'haha') milein, ya 15 letter se chota message ho, ya photo mobile ka screenshot ho → AI turant use **Fake / Spam** mark kar deta hai.
- **2. Zameen Check (Terrain Physics):** Cuttack jaisi flat zameen par agar koi 'Landslide' report karega, toh AI use reject kar dega kyunki flat zameen par pahad nahi hote.
- **3. Points Dena (Scoring):** Bada detailed message (+25 pts), GPS fix (+15 pts), asli camera photo (+20 pts), mausam match (+20 pts) se total score banta hai.
- **4. Faisla (Decision Thresholds):** **Score 70% se zyada → Genuine (Asli), 40% se 70% → Needs Review, 40% se kam → Fake / Suppressed.**

🔍 Evaluator Pitch: Why this model & Why not heavy LLMs (GPT-4 / Cloud CNN)?

"Sir, disaster me mobile towers aur internet band ho jate hain. Agar hum heavy GPT-4 lagate, toh bina internet ke app fail ho jata aur API billing cost lagti. Hamara in-browser agent client-side CPU par 1 millisecond me bina internet ke 100% offline fake reports filter kar deta hai."

2.0 LIVE TELEMETRY & MULTI-HAZARD SENSOR INGESTION

MODULE 2 • ALERTS PIPELINE

Q2 In the Live Alerts section, how are earthquakes and floods detected, what algorithm is used, and which APIs are integrated?

MODEL / PROCESSOR

Telemetry Anomaly Detector (Wavefront & Meteorological Processor)

DATA FEEDS & APIS

USGS Seismic GeoJSON API, Open-Meteo Weather REST API

⚙️ Detection Thresholds & Algorithm Logic:

- **Earthquake (Bhukamp):** USGS feed se live Richter scale magnitude dekhta hai: **Magnitude 5.5 se zyada → Critical Red Alert, 4.0 se 5.4 → Warning Yellow Alert.** Sath hi kitne KM tak asar hoga (Impact Radius) nikal kar map par circle bana deta hai.
- **Flood (Baadh) & Toofan:** Open-Meteo Doppler satellite se barish aur hawa check karta hai: **Barish 5 mm/hour se zyada → Flash Flood Warning, Hawa 35 km/hour se zyada → Cyclone / Gale Alert.**

🔍 Evaluator Pitch: Why these APIs & Why not commercial paid weather APIs?

"Sir, USGS aur Open-Meteo authoritative open-science APIs hain jo 100% free hain, inme koi rate-limit nahi hoti, aur yeh ground sensors ka live physical data deti hain jisse false alarms eliminate ho jate hain."

3.0 BAYESIAN RISK DENSITY & MULTI-HAZARD RADAR ENGINE

MODULE 3 • RISK FORECAST

Q3 In the Dashboard & Risk Forecast section, how is disaster risk probability calculated, and why use this model instead of simple percentages?

MATHEMATICAL MODEL

Bayesian Bell Curve (Gaussian Density) + 6-Axis Radar Normalizer

VISUALIZATION ENGINE

Open-Meteo Telemetry + Recharts GPU SVG Canvas

⚙️ Probability Calculation Rules:

- **Bell Curve Graph:** Live mausam ke hisaab se curve ka peak aage-picche shift hota hai. Normal mausam me curve **Low Risk (20-25%)** par rehta hai, lekin barish aur hawa badhne par curve ka peak **High Risk (75-85%)** par shift ho jata hai.
- **Radar Chart:** Ek sath 6 khatron (Flood, Cyclone, Heatwave, Earthquake, Drought, Tsunami) ka live 0 se 100% risk meter ek hi hexagon graph me render karta hai.

🔑 Evaluator Pitch: Why this model & Why not simple static percentages?

"Sir, sirf '60% chance' bolne se disaster ka exact khatra samajh nahi aata. Bell Curve graph citizen aur NDRF commanders ko poori probability spread aur sensor confidence dikhata hai."

Q4 In the Map & Navigation section, how does shortest safe path routing work, which algorithm is used, and what happens when internet goes down?

ROUTING ALGORITHMS
Dijkstra / OSRM Graph Routing, Haversine Distance, Sinusoidal
Offline Vector

GIS APIS USED
Project OSRM Engine, Komoot Photon OSM API, Leaflet Map

-  **Navigation & Pathfinding Logic:**
- **Shortest Safe Path: Dijkstra / OSRM Algorithm** road network par flooded aur blocked raste ko bypass karke sabse chhota aur safe road rasta nikalta hai.
 - **Distance Calculation: Haversine Algorithm** GPS coordinates se exact curved kilometer distance calculate karta hai.
 - **Offline Tactical Route:** Agar internet band ho, toh hamara offline curve formula 32 km/h emergency speed ke hisaab se safe rasta aur ETA nikal deta hai.

 **Evaluator Pitch: Why Leaflet & OSRM instead of Google Maps API?**
"Sir, Google Maps API mehenge billing charges lagati hai aur offline caching allow nahi karti. Leaflet aur OSRM 100% free hain, open-source hain aur flooded khatron ko bypass karke offline safe path nikalte hain."

Q5 In the Emergency Help section, how are nearby hospitals and shelters found, and how does 1-tap SOS work?

SEARCH ALGORITHM
Bounding Box Spatial POI Query + Proximity Sorting

SOS PROTOCOLS
Native Tel URI Protocol (112, 108, 101), Web Share API

-  **Facility Discovery & SOS Calling Logic:**
- **Facility Discovery:** User ke GPS se 15-45km radius me hospitals, police thana, fire brigade, aur shelters dhoondh kar sabse paas wale ko top par sort karta hai.
 - **1-Tap SOS Protocol:** Direct phone ka dialer trigger karta hai (Police: 112, Ambulance: 108, Fire: 101) aur sath hi WhatsApp/SMS par victim ke exact GPS coordinates share kar deta hai.

 **Evaluator Pitch: Why direct hardware protocol?**
"Sir, disaster me victim ghabraya hota hai aur type nahi kar sakta. 1-Tap direct hardware call lagata hai aur bina internet ke location broadcast kar deta hai."

Q6 How does volunteer incident triage work, and how does the app persist and sync data during complete network blackouts?

TRIAGE & LIFECYCLE
Priority Triage Sorting + 6-Step Mission Lifecycle State Machine

STORAGE ARCHITECTURE
IndexedDB v2 (5 Stores) + FIFO Offline Sync Queue Manager

-  **Triage & Sync Reconciliation Workflow:**
- **Volunteer Triage:** Sabhi fake/spam reports suppress ho jati hain. Ranking rules: **Critical (Jaan ka khatra) > Medium > Nearest Proximity.**
 - **Offline Persistence:** Internet na hone par report phone ke IndexedDB me PendingSync tag ke sath save ho jati hai. Internet aate hi OfflineSyncManager background me auto-upload kar deta hai.

 **Evaluator Pitch: Why IndexedDB why not LocalStorage?**
"Sir, LocalStorage sirf 5MB data rakh sakta hai aur phone ko hang kar deta hai. IndexedDB 500MB+ data aur photos ko bina kisi UI lag ke offline store kar sakta hai."

SECTION / MODULE	MODEL / METHOD	ALGORITHM RULES	WHY THIS MODEL? (PUNCHLINE)
1. Report Section	Bayesian NLP + Image Checker	Score $\geq$ 70% Genuine, < 40% Fake/Spam	Phone CPU par 1ms me bina internet chalta hai.
2. Live Alerts	Wavefront & Anomaly Detector	USGS Richter $\geq$ 5.5 Critical, Rain $\geq$ 5mm Flood	Real government sensor data, zero false alarms.
3. Risk Probability	Bayesian Bell Curve + Radar	Live Weather Shift (Low 25% $\rightarrow$ High 85%)	Static number ke badle complete risk spread deta hai.
4. Disaster Map	OSRM Dijkstra + Sinusoidal	Haversine Distance + Flooded Obstacle Bypass	100% Free, open-source aur offline safe rasta deta hai.
5. Volunteer Desk	Priority Triage + 6-Step Flow	Critical > Medium > Nearest Proximity	Fake calls rokta hai aur duplicate rescue bachata hai.
6. Offline Sync	IndexedDB + Sync Queue	FIFO Auto Reconnection Sync	500MB+ offline storage deta hai bina phone hang kiye.